



Ahsanullah University of Science & Technology

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

Project name: Miniature Uninterruptible Power Supply (UPS)

Course no: EEE 4154

Course title: Power System-II lab

Group No: 02

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Project Title:

Miniature uninterruptible power supply (UPS)

Objective:

To deliver continuous AC power to a connected load without even the slightest interruption, regardless of unforeseen power fluctuations or brownouts.

Equipment:

Serial No	Equipment Name	Quantity
01	Transformer 12V	2x
02	Diode	3x
03	Buck Converter	1x
04	Fixed Resistors (1K)	3x
05	Variable Resistors (10K)	1x
06	Capacitors (1000 μ F, 104pF)	1x,1x
07	MOSFETs (IRF3205)	2x
08	IC CD4047	1x
09	Battery 12V (4V LEAD ACID BATTERY)	3x
10	Relay	1x
11	Switch	1x
12	2 Pin Power Cord	1x
13	Voltmeter	1x

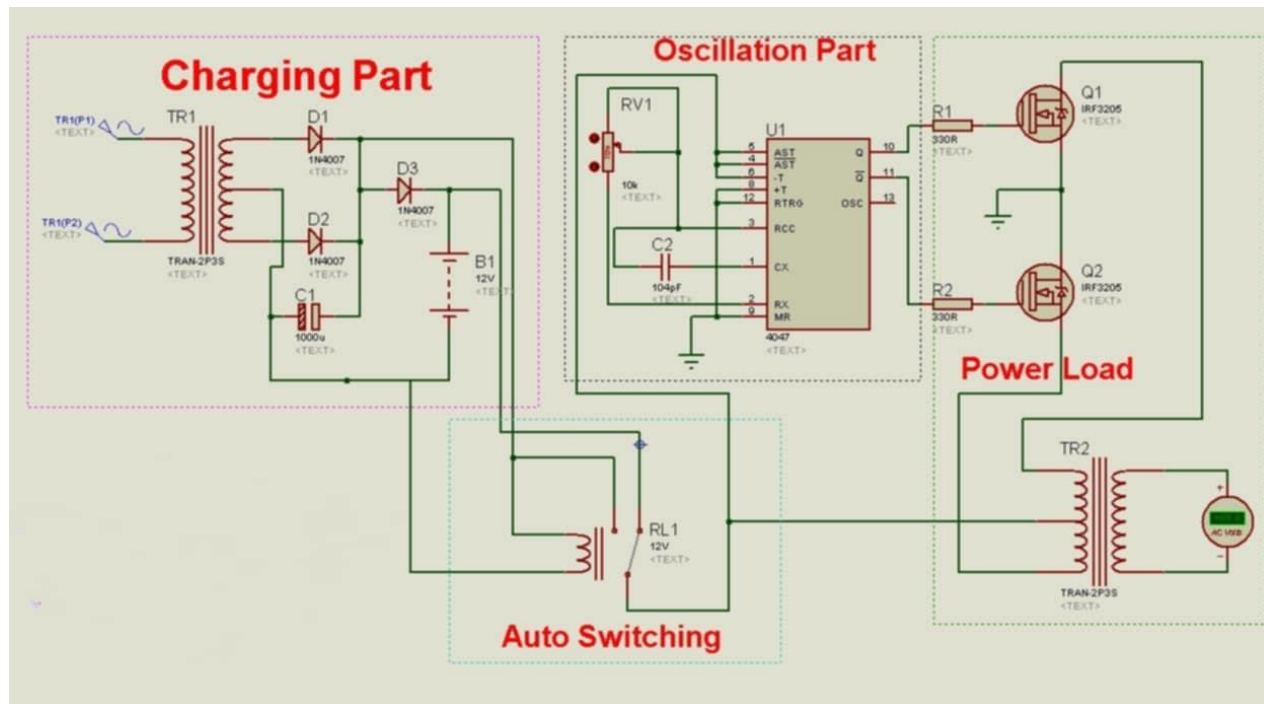
Circuit Diagram: (Proteus view)

Figure: Schematic of UPS in PROTEUS

Circuit Diagram: (Physical view)

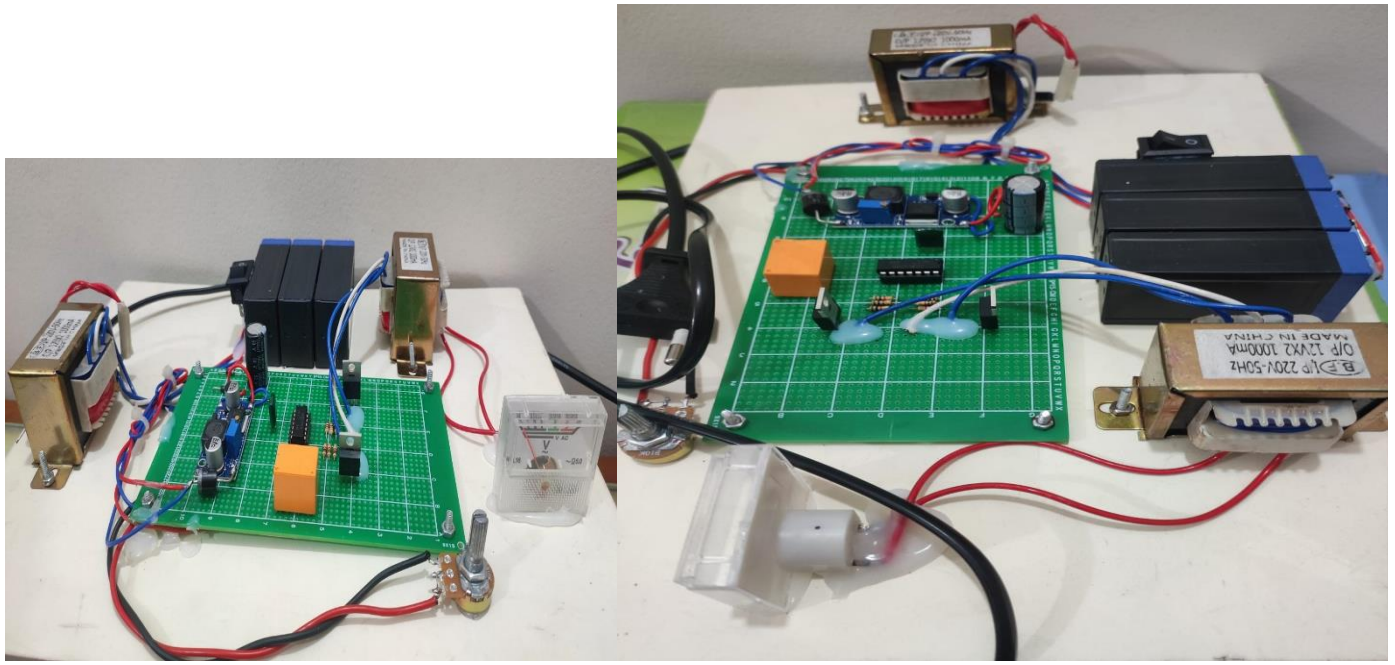


Figure: Physical view of the UPS

Working Principle:

The working principle of this circuit can be divided into 4 parts which are described below:

1. Charging Part:

- **Function:** Converts AC to DC and stores energy in a 12V battery.
- **Working:**
 - The transformer **TR1** steps down the AC voltage to a lower voltage.
 - The diodes **D1, D2, and D3** form a **bridge rectifier**, converting AC to pulsating DC.
 - The capacitor **C1 (1000 μ F)** smooths the DC voltage.
 - The 12V **battery B1** stores the charge for use in the inverter circuit.

2. Oscillation Part:

- **Function:** Generates an oscillating signal to drive MOSFETs in the power stage.
- **Working:**
 - The IC **4047** is configured as an **Astable Multivibrator**, producing a **square wave signal**.
 - The resistor **RV1 (10k Ω)** and capacitor **C2 (104pF)** set the oscillation frequency.
 - The IC outputs two complementary square wave signals at **pins 10 and 11**, which control the MOSFETs.

3. Auto Switching:

- **Function:** Automatically switches the inverter ON or OFF based on battery status.
- **Working:**
 - The **relay (RL1, 12V)** controls power to the oscillation and power load sections.
 - It ensures the inverter only operates when needed, preventing over-discharge of the battery.

4. Power Load:

- **Function:** Converts DC from the battery into AC and steps up the voltage.
- **Working:**
 - The square wave signals from **IC 4047** drive the **MOSFETs (Q1 & Q2, IRF3205)**.
 - The MOSFETs act as high-speed switches, alternately switching ON and OFF, allowing current to flow through **transformer TR2**.
 - **Transformer TR2** steps up the 12V AC to a higher AC voltage.
 - The **AC meter** measures the output AC voltage.

Final Output:

- The **DC from the battery is converted into AC** using the oscillation and switching process.
- The **AC voltage is stepped up** using **TR2**, providing usable **AC output**.

This circuit is useful for **powering AC devices from a 12V DC source**, commonly used in **UPS systems, solar inverters, and emergency power supplies**.

Applications of UPS:

- **IT & Computers:** Protects systems, prevents data loss, and ensures safe shutdown.
- **Medical:** Supports critical equipment in hospitals.
- **Telecommunications:** Maintains network operations.
- **Industry:** Prevents downtime for automation and assembly lines.
- **Banking:** Powers ATMs and financial systems.
- **Home:** Keeps essential devices running during outages.
- **Security:** Ensures uninterrupted CCTV and alarm operation.
- **Education:** Powers teaching aids like projectors and computers.
- **Research:** Protects lab equipment and experiment data.
- **Emergency Services:** Supports disaster management systems.

Advantages:

- **Simple and cost-effective design** using commonly available components.
- **Efficient MOSFET-based switching** ensures minimal power loss.
- **Auto-switching relay** allows uninterrupted operation between charging and inverter mode.
- **Adjustable frequency control** via the oscillator circuit.

Limitations and Possible Improvements:

- ♦ **Square wave output** may cause noise and inefficiencies in some appliances. A **PWM-based or sine wave inverter design** would improve compatibility.
- ♦ **MOSFET heating issues** may arise at higher loads, requiring **heat sinks or cooling mechanisms**.
- ♦ **Battery capacity limitations** may affect runtime; using a **higher Ah battery** can extend usage.
- ♦ **Transformer selection** impacts efficiency. Using a **high-frequency ferrite-core transformer** can reduce losses.

Conclusion

This inverter circuit effectively converts 12V DC into AC voltage using a square wave approach. It is ideal for basic power backup applications but can be enhanced with PWM control and sine wave output for improved performance. Future upgrades could include microcontroller-based control and better output filtering to optimize efficiency and compatibility with various electronic loads.